

Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

What is the purpose of this section?

- Context and importance of your research.
- Outline of literatur review

Why choose image processing techniques?

- Former methods cost too much time and were expensive + destructive [1]
- Focus on image analysis component [2]
- Review plant phenotyping concepts in relation to image processing and plant breeding, discussing various measurable traits

1 Plant Phenotyping

Plant breeding is defined as the process of inducing desirable inheritable traits in plants through controlled human action [3], which is important in solving the problem of providing an adequate global food supply. The rapid growth of the human population has created a strain on international food security. It is estimated that crop production must increase by 70% by 2050 [4] to meet this demand. Additionally, abiotic and biotic stressors limit cultivar biomass and yield [5]. Therefore, plant breeders need effective techniques to produce germplasm for cultivars that maximize yield and biomass, in various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have been an effective tool for plant breeding [6].

1.1 What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [7], where phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [8]. Phenotypes can alternatively be seen as a combination of all morphological, physiological, chemical, and behavioural aspects that represent the individual plant [9, 10, 11].

1.2 Plant Phenotyping As Method For Improving Plant Traits

Quantitative data produced by plant phenotyping is most useful in analyzing plant-environment interactions in plant breeding [7] for the development of germplasm with improved plant traits [11]. The data produced by phenotyping is used as a proxy for assessing a plant's performance in relation to its environment. For example, stressors or disturbances of plants are induced by environmental conditions that organisms are placed in, and these stressors induce symptoms which can be detected and evaluated using phenotyping. This indicates phenotypical traits of a plant have a complex dynamic feedback loop with their environment [12]. Therefore phenotypical traits serve as proxies in determining whether certain genotypes have favourable traits in particular environmental conditions.

1.3 Image-Obtainable Phenotypical Traits As Indicators Of Plant Health, Growth, And Stress

A phenotypical trait is classifiable as morphological, physiological, or temporal. Additionally, the literature indicates morphological and physiological phenotypical traits are classifiable as holistic-based or component-based [13]. Holistic properties consider the plant as a whole, whereas component-based features look at organs like the organism's fruit, flower, leaves, stem, etc.

Leaf-derived traits are one of the most commonly used plant status indicators because leaves rapidly respond to environmental stimuli and robustly indicate various factors that determine plant health, growth and stress. [14]. Traits of leaves can monitor and predict a host of plant health, growth and stress factors. For example, leaf area predicts plant productivity because reduced leaf area results in decreased chlorophyll content and less photosynthesis. Therefore, leaf area is a good proxy measure for evaluating plant growth and has been used to monitor and estimate crop yield and health [15]. Various other leaf morphological traits exist in plant phenotyping. These include leaf number and inclination angle for monitoring plant growth and development [16, 14], and leaf thickness for measuring abiotic stressors like water scarcity [17]. Additionally, physiological traits of leaves exist that indicate various factors of plant status like stomatal conductance for the detection of biotic stressors like pathogens [18], leaf water content for drought stress [19].

Although leaves play an important role in evaluating plant growth, health and resilience to stress, these evaluations cannot be scaled to the whole plant [14]. Rather, other parts of the plant should be phenotyped to understand the processes governing plant status comprehensively. The literature is rife with such examples that study traits of roots [20], stems [21], seeds [22], and fruit [23].

2 Image Processing And Plant Phenotyping

Traditional plant phenotyping methods are destructive, prone to human error, time-consuming, and challenging to perform from start to end of a plant's ontogenesis [24, 14]. Recent advances in image sensing, processing, and data management have produced high-throughput image-based plant phenotyping systems, which are amenable to automation and can analyse large samples of data precisely in a very short period. Reducing the need for potentially erroneous, time-consuming, expensive human labour in plant phenotyping [13].

These Modern high-throughput image-based plant phenotyping systems are examples of cyberinfrastructure (CI) systems [23], which are research environments that support data acquisition, storage, management and computing and processing services distributed over some network [25]. For image-based plant phenotyping, this means that image acquisition and processing techniques are vital.

2.1 Image Acquisition Techniques In Plant Phenotyping Systems

Image processing techniques in plant phenotyping cannot be applied until suitable images are acquired. Image sensor advancements have provided the foundations for detecting various information from plants using electromagnetic radiation. Plant tissues interact with this radiation by absorbing, reflecting, emitting, transmitting, and fluorescing light. Different ranges of wavelengths correspond to different biochemical or physiological aspects of plant tissue, which are not visible to the naked eye [11]. Thus, the choice of imaging technique for data acquisition affects the types of phenotypic parameters that can be measured.

Current imaging techniques are visible, fluorescence, thermal, tomographic, and hyperspectral imaging [26, 11], where the most commonly used techniques are visible, and hyperspectral imaging due to their affordability, robustness and flexibility in measuring a wide range of phenotypic parameters.

The most common imaging technique is visible-imaging. Typically affordable, and high-resolution RGB cameras are used in visible imaging. These devices are sensitive to the visible spectral range and have been used to measure various phenotypic parameters like shoot biomass, yield and leaf morphology, and salinity tolerance [27, 28].

Hyperspectral imaging is another widely used in plant phenotyping. In this technique, sensors capture multiple spectral ranges, enabling detailed measurements for various phenotypic parameters. Notable applications of hyper-

spectral imaging include estimating nutrient content and detecting plant responses to abiotic or biotic stressors [29].

The other imaging techniques are not as widely used as visible or hyperspectral imaging; however, they are also effective in measuring plant phenotypic parameters. For instance, thermal imaging in plant water stress relations [11, 18], fluorescence imaging in used plant pathogenic symptom monitoring [30, 31], and tomography in estimating plant organ water content [32].

2.2 Image Processing Techniques In Plant Phenotyping Systems

There is a lack of generalised or standardised image-processing techniques in plant phenotyping; hence, bespoke image-processing pipelines are designed for specific plant species or genotypes like *Arabidopsis* [11, 33]. Therefore, various phenotyping systems' image processing and analysis pipelines can vastly differ; however, as mentioned earlier in the review of plant phenotyping traits, various morphological traits are derived from specific organs or parts of the plant. Furthermore, segmentation methods directly affect the accuracy of more complex tasks like object detection, 3D reconstruction, and classification. Hence, the validity and preciseness of the derived phenotypical measurements and analysis depend on the image segmentation method selected [34, 35].

In plant phenotyping, appropriate selection of segmentation techniques is vital when considering the type of phenotypic parameter to be measured. Segmentation methods in plant phenotyping are typically classifiable as traditional or learning-based [34]. Traditional segmentation methods in plant phenotyping typically target holistic phenotypical traits like plant height and include frame differencing [36], colour-based [37], edge detection [38], spectral band difference-based [39], shape modelling-based [40], and graph-based segmentation [41]. In contrast to traditional techniques, learning-based methods are more suitable for targeting component-based traits of specific organs or parts of a plant [34]. Many learning-based methods use deep learning and machine-learning mechanisms to handle segmentation [42, 43].

What is the purpose of this section?

- Discuss how image processing and plant phenotyping are applied in low-cost automated greenhouses
- What methods, limitations and advances have there been.
- What does a typical image preprocessing pipeline look like?

Basis for future in low-cost imaging analysis

The basis for the points below is from [25]

- Existing phenotyping systems are commercially developed and under patent, so are not modifiable or readily available [44]
- RGB, and hyperspectral sensors most commonly used as they are affordable
- Data collection and analysis tools are not generalizable or standardised, so many custom flows are made
- Indicating the need for standardised, extendable image analysis pipelines using readily available open-source software and hardware
- Recent advances show the use of cheap hardware like raspberry pi, and affordable RGB sensors [45]
- However these sensors don't have a lot of compute power, so analysis and preprocessing is done remotely
- Cheap sensors but are cheaper and faster
- Cloud architectures exist, but with amount of data generated this cost can be prohibitive in terms of processing cost [46]. Cyberinfrastructure expensive ref (95,96)
- Moreover, existing systems are not designed for multiple genotypes so their applicability is limited, suggesting need for more generalizable image analysis
- Deep learning and machine learning can be used to automate image analysis, and reduce costs
- Availability of large labelled data is bottleneck
- To use on cheap hardware you also need to optimize
- Therefore low complexity efficient algorithms for automated image preprocessing and image analysis is needed
- Using multimodal segmentation is also another angle

- ref[36], one of the advantages of using ML approaches in plant phenotyping is their ability to search large datasets and discover patterns by simultaneously looking at a combination of features (compared to analyzing each feature separately). This was previously a challenge because of the high dimensionality of plant images and their large quantity, making them difficult to analyze through traditional processing methods (ref 37)
- machine learning achieved using supervised learning, so we aren't able to detect novel or unexpected traits using unsupervised learning (ref 40)
- traditional methods in image processing not generalizable, in pathogen detection and require strict conditions and custom workflows to work [24]
- Current research Focus on image sensing, rather on computation which is new bottleneck. Imaging technology has advanced, but there is lack of robust analysis and computation tools. [46]
- Sensors generate huge amount of data, therefore algorithms need to be designed to deal with Terabytes of data. Furthermore, low cost computing units have limited bandwidth, battery and sensor capabilities indicating that algorithms need to take this in to account. Low complexity and efficient algorithms that can run on resource constrained tech is needed [46]
- Mobile applications can be used and run processing without need for external network, cyberinfrastructure. However there are not many applications available for this [35]. Image analysis tools are often too expensive to run on these devices due to the constrained resources, however they are much cheaper and therefore can be exploited. In particular low complexity efficient systems, which generalize to various genotypes, and phenotypic measurements are needed. In particular the development of efficient low resource intensive machine modelling and deep learning techniques can be made.
- preprocessing technique choice is also needed, for example image segmentation methods can be costly, and a suitable method depends on various factors. Certain methods have cheaper cost, but less accuracy [35]. In developing low-cost image analysis and preprocessing tools for phenotyping it is necessary to weight the cost savings versus accuracy and precision. But bespoke image processing techniques don't lend well to generalizability and extendability.
- cost-effective solutions require carefully choosing sensors and data processing techniques to be most cost-effective [14]
- automated systems not built for general solutions but for specific species of plants [7]

Current applications

- Often combined with machine learning algorithms
- Cheap Raspberry PI with RGB camera [45]
- Autonomous robots with sensors and cameras [47]
- use off the shelf commercial software like microsoft xbox360 kinect [48]

What affects a low-cost automated greenhouse cost

Points from this paper [49]

- Image preprocessing pipeline
- open source software
- hardware
- amount of labour and time needed for analysis
- energy cost of technology
- amount of data generated, where is it stored, how frequently? [2]

Challenges of low-cost automated greenhouses

- Many phenotyping systems existing are commercially developed under patent so hardware and software can not be modified. [44]
- Current applications use moving conveyor belt (benchtop) [44]
- or conveyer belt [44]
- or moving robotic arm to move sensors around to get high-throughput measurements.
- Robot solutions not cheap for farmers [47]
- large amounts of data need to be collected, and processed. Existing phenotyping platforms exist, but they are expensive and have restricted running environments.
- Many systems developed for laboratory systems, but not extended for real applications. Standardised off the shelf equipment, open-source software allow modular and extensible plus affordable solutions [50]
- Custom data collection schemes need to be made very little standards
- lack of robust methods to analyse plant images
- low cost moving track systems limited in phenotypical, genotype and management types are supported [51] suggesting need for low-cost systems that can support a wider variety of genotypes and phenotypical traits.

Fill out the conclusion section.

References

- [1] R. H. Laxman et al. “Non-Invasive Quantification of Tomato (*Solanum Lycopersicum* L.) Plant Biomass through Digital Imaging Using Phenomics Platform”. In: *Indian Journal of Plant Physiology* 23.2 (June 1, 2018), pp. 369–375. ISSN: 0974-0252. DOI: 10.1007/s40502-018-0374-8. URL: <https://doi.org/10.1007/s40502-018-0374-8> (visited on 05/24/2025).
- [2] Jianjun Du et al. “Greenhouse-Based Vegetable High-Throughput Phenotyping Platform and Trait Evaluation for Large-Scale Lettuces”. In: *Computers and Electronics in Agriculture* 186 (July 2021), p. 106193. ISSN: 01681699. DOI: 10.1016/j.compag.2021.106193. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0168169921002106> (visited on 05/24/2025).
- [3] Thomas J. Orton. “Introduction”. In: *Horticultural Plant Breeding*. Elsevier, 2020, pp. 3–7. ISBN: 978-0-12-815396-3. DOI: 10.1016/B978-0-12-815396-3.09986-8. URL: <https://linkinghub.elsevier.com/retrieve/pii/B9780128153963099868> (visited on 05/16/2025).
- [4] M. Carvajal-Yepes et al. “A Global Surveillance System for Crop Diseases”. In: *Science* 364.6447 (June 28, 2019), pp. 1237–1239. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.aaw1572. URL: <https://www.science.org/doi/10.1126/science.aaw1572> (visited on 05/21/2025).
- [5] J.P. Grime. “Evidence for the Existence of Three Primary Strategies in Plants and Its Relevance to Ecological and Evolutionary Theory”. In: *The American Naturalist* 111.982 (1997), p. 1169. DOI: 10.1086/283244. URL: <https://www.journals.uchicago.edu/doi/abs/10.1086/283244> (visited on 05/19/2025).
- [6] Achala Anand, Madhumitha Subramanian, and Debasish Kar. “Breeding Techniques to Dispense Higher Genetic Gains”. In: *Frontiers in Plant Science* 13 (Jan. 19, 2023). ISSN: 1664-462X. DOI: 10.3389/fpls.2022.1076094. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2022.1076094/full> (visited on 05/16/2025).
- [7] Fabio Fiorani and Ulrich Schurr. “Future Scenarios for Plant Phenotyping”. In: *Annual Review of Plant Biology* 64.1 (Apr. 29, 2013), pp. 267–291. ISSN: 1543-5008, 1545-2123. DOI: 10.1146/annurev-arplant-050312-120137. URL: <https://www.annualreviews.org/doi/10.1146/annurev-arplant-050312-120137> (visited on 05/14/2025).
- [8] W. Johannsen. “The Genotype Conception of Heredity”. In: *The American Naturalist* 45.531 (Mar. 1911), pp. 129–159. ISSN: 0003-0147, 1537-5323. DOI: 10.1086/279202. URL: <https://www.journals.uchicago.edu/doi/10.1086/279202> (visited on 05/14/2025).

- [9] Qiaosheng Guo and Zaibiao Zhu. “Phenotyping of Plants”. In: *Encyclopedia of Analytical Chemistry*. John Wiley & Sons, Ltd, 2014, pp. 1–15. ISBN: 978-0-470-02731-8. DOI: 10.1002/9780470027318.a9934. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470027318.a9934> (visited on 05/14/2025).
- [10] Semira Ebrahim Mohammed. “Review on Plant Phenotyping”. In: (). URL: <https://www.abrinternationaljournal.org/abstract/review-on-plant-phenotyping-1102322.html> (visited on 05/05/2025).
- [11] Lei Li, Qin Zhang, and Danfeng Huang. “A Review of Imaging Techniques for Plant Phenotyping”. In: *Sensors (Basel, Switzerland)* 14.11 (Oct. 24, 2014), pp. 20078–20111. ISSN: 1424-8220. DOI: 10.3390/s141120078. PMID: 25347588. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4279472/> (visited on 04/20/2025).
- [12] Achim Walter, Frank Liebisch, and Andreas Hund. “Plant Phenotyping: From Bean Weighing to Image Analysis”. In: *Plant Methods* 11.1 (Mar. 4, 2015), p. 14. ISSN: 1746-4811. DOI: 10.1186/s13007-015-0056-8. URL: <https://doi.org/10.1186/s13007-015-0056-8> (visited on 05/05/2025).
- [13] Sruti Das Choudhury, Ashok Samal, and Tala Awada. “Leveraging Image Analysis for High-Throughput Plant Phenotyping”. In: *Frontiers in Plant Science* 10 (Apr. 24, 2019). ISSN: 1664-462X. DOI: 10.3389/fpls.2019.00508. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2019.00508/full> (visited on 05/21/2025).
- [14] Huichun Zhang et al. “High-Throughput Phenotyping of Plant Leaf Morphological, Physiological, and Biochemical Traits on Multiple Scales Using Optical Sensing”. In: *The Crop Journal* 11.5 (Oct. 1, 2023), pp. 1303–1318. ISSN: 2214-5141. DOI: 10.1016/j.cj.2023.04.014. URL: <https://www.sciencedirect.com/science/article/pii/S2214514123000740> (visited on 05/20/2025).
- [15] Juanjuan Zhang et al. “Leaf Area Index Estimation Model for UAV Image Hyperspectral Data Based on Wavelength Variable Selection and Machine Learning Methods”. In: *Plant Methods* 17.1 (May 3, 2021), p. 49. ISSN: 1746-4811. DOI: 10.1186/s13007-021-00750-5. URL: <https://doi.org/10.1186/s13007-021-00750-5> (visited on 05/25/2025).
- [16] J. Praveen Kumar and S. Domnic. “Rosette Plant Segmentation with Leaf Count Using Orthogonal Transform and Deep Convolutional Neural Network”. In: *Machine Vision and Applications* 31.1–2 (Feb. 2020), p. 6. ISSN: 0932-8092, 1432-1769. DOI: 10.1007/s00138-019-01056-2. URL: <http://link.springer.com/10.1007/s00138-019-01056-2> (visited on 05/25/2025).

- [17] Amin Afzal, Sjoerd W. Duiker, and John E. Watson. “Leaf Thickness to Predict Plant Water Status”. In: *Biosystems Engineering* 156 (Apr. 1, 2017), pp. 148–156. ISSN: 1537-5110. DOI: 10.1016/j.biosystemseng.2017.01.011. URL: <http://www.scopus.com/inward/record.url?scp=85013072400&partnerID=8YFLogxK> (visited on 05/25/2025).
- [18] Roselyne Ishimwe, K. Abutaleb, and F. Ahmed. “Applications of Thermal Imaging in Agriculture—A Review”. In: *Advances in Remote Sensing* 03.03 (2014), pp. 128–140. ISSN: 2169-267X, 2169-2688. DOI: 10.4236/ars.2014.33011. URL: <http://www.scirp.org/journal/doi.aspx?DOI=10.4236/ars.2014.33011> (visited on 05/19/2025).
- [19] Kenta Watanabe et al. “Fundamental Study on Water Stress Detection in Sugarcane Using Thermal Image Combined with Photosynthesis Measurement Under a Greenhouse Condition”. In: *Sugar Tech* 24.5 (Oct. 1, 2022), pp. 1382–1390. ISSN: 0974-0740. DOI: 10.1007/s12355-021-01087-y. URL: <https://doi.org/10.1007/s12355-021-01087-y> (visited on 05/25/2025).
- [20] Gernot Bodner et al. “Hyperspectral Imaging: A Novel Approach for Plant Root Phenotyping”. In: *Plant Methods* 14.1 (Oct. 3, 2018), p. 84. ISSN: 1746-4811. DOI: 10.1186/s13007-018-0352-1. URL: <https://doi.org/10.1186/s13007-018-0352-1> (visited on 05/25/2025).
- [21] Sruti Das Choudhury et al. “Automated Stem Angle Determination for Temporal Plant Phenotyping Analysis”. In: *Proceedings of the IEEE International Conference on Computer Vision Workshops*. 2017, pp. 2022–2029. URL: https://openaccess.thecvf.com/content_ICCV_2017_workshops/w29/html/Choudhury_Automated_Stem_Angle_ICCV_2017_paper.html (visited on 05/25/2025).
- [22] Chongyuan Zhang et al. “High-Throughput Phenotyping of Seed/Seedling Evaluation Using Digital Image Analysis”. In: *Agronomy* 8.5 (5 May 2018), p. 63. ISSN: 2073-4395. DOI: 10.3390/agronomy8050063. URL: <https://www.mdpi.com/2073-4395/8/5/63> (visited on 05/25/2025).
- [23] Alebel Mekuriaw Abebe et al. “Image-Based High-Throughput Phenotyping in Horticultural Crops”. In: *Plants* 12.10 (10 Jan. 2023), p. 2061. ISSN: 2223-7747. DOI: 10.3390/plants12102061. URL: <https://www.mdpi.com/2223-7747/12/10/2061> (visited on 05/25/2025).
- [24] Jayme Garcia Arnal Barbedo. “Digital Image Processing Techniques for Detecting, Quantifying and Classifying Plant Diseases”. In: *SpringerPlus* 2.1 (Dec. 7, 2013), p. 660. ISSN: 2193-1801. DOI: 10.1186/2193-1801-2-660. URL: <https://doi.org/10.1186/2193-1801-2-660> (visited on 05/11/2025).
- [25] Shona Nabwire et al. “Review: Application of Artificial Intelligence in Phenomics”. In: *Sensors* 21.13 (June 25, 2021), p. 4363. ISSN: 1424-8220. DOI: 10.3390/s21134363. URL: <https://www.mdpi.com/1424-8220/21/13/4363> (visited on 05/25/2025).

- [26] *Sensors for Measuring Plant Phenotyping: A Review*. URL: <http://www.ijabe.net/cn/article/doi/10.25165/j.ijabe.20181102.2696> (visited on 05/09/2025).
- [27] Mahmood R Golzarian et al. “Accurate Inference of Shoot Biomass from High-Throughput Images of Cereal Plants”. In: *Plant Methods* 7.1 (Dec. 2011), p. 2. ISSN: 1746-4811. DOI: 10.1186/1746-4811-7-2. URL: <https://plantmethods.biomedcentral.com/articles/10.1186/1746-4811-7-2> (visited on 05/26/2025).
- [28] V. Hoyos-Villegas et al. “Ground-Based Digital Imaging as a Tool to Assess Soybean Growth and Yield”. In: *Crop Science* 54.4 (2014), pp. 1756–1768. ISSN: 1435-0653. DOI: 10.2135/cropsci2013.08.0540. URL: <https://onlinelibrary.wiley.com/doi/abs/10.2135/cropsci2013.08.0540> (visited on 05/26/2025).
- [29] Rijad Sarić et al. “Applications of Hyperspectral Imaging in Plant Phenotyping”. In: *Trends in Plant Science* 27.3 (Mar. 1, 2022), pp. 301–315. ISSN: 1360-1385. DOI: 10.1016/j.tplants.2021.12.003. URL: <https://www.sciencedirect.com/science/article/pii/S1360138521003460> (visited on 05/22/2025).
- [30] Hartmut K. Lichtenthaler and Ursula and Rinderle. “The Role of Chlorophyll Fluorescence in The Detection of Stress Conditions in Plants”. In: *C R C Critical Reviews in Analytical Chemistry* 19 (supl Jan. 1, 1988), S29–S85. ISSN: 0007-8980. DOI: 10.1080/15476510.1988.10401466. URL: <https://doi.org/10.1080/15476510.1988.10401466> (visited on 05/22/2025).
- [31] Philip J. Swarbrick, Paul Schulze-Lefert, and Julie D. Scholes. “Metabolic Consequences of Susceptibility and Resistance (Race-Specific and Broad-Spectrum) in Barley Leaves Challenged with Powdery Mildew”. In: *Plant, Cell & Environment* 29.6 (2006), pp. 1061–1076. ISSN: 1365-3040. DOI: 10.1111/j.1365-3040.2005.01472.x. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-3040.2005.01472.x> (visited on 05/26/2025).
- [32] Carel W. Windt et al. “MRI of Long-Distance Water Transport: A Comparison of the Phloem and Xylem Flow Characteristics and Dynamics in Poplar, Castor Bean, Tomato and Tobacco”. In: *Plant, Cell & Environment* 29.9 (2006), pp. 1715–1729. ISSN: 1365-3040. DOI: 10.1111/j.1365-3040.2006.01544.x. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1365-3040.2006.01544.x> (visited on 05/26/2025).
- [33] Samuel Arvidsson, Paulino Pérez-Rodríguez, and Bernd Mueller-Roeber. “A Growth Phenotyping Pipeline for Arabidopsis Thaliana Integrating Image Analysis and Rosette Area Modeling for Robust Quantification of Genotype Effects”. In: *New Phytologist* 191.3 (2011), pp. 895–907. ISSN: 1469-8137. DOI: 10.1111/j.1469-8137.2011.03756.x. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1469-8137.2011.03756.x> (visited on 05/26/2025).

- [34] Sruti Das Choudhury. “Segmentation Techniques and Challenges in Plant Phenotyping”. In: *Intelligent Image Analysis for Plant Phenotyping*. 1st ed. CRC Press, 2020. ISBN: 978-1-315-17730-4.
- [35] Mark Müller-Linow et al. “Plant Screen Mobile: An Open-Source Mobile Device App for Plant Trait Analysis”. In: *Plant Methods* 15.1 (Dec. 2019), p. 2. ISSN: 1746-4811. DOI: 10.1186/s13007-019-0386-z. URL: <https://plantmethods.biomedcentral.com/articles/10.1186/s13007-019-0386-z> (visited on 05/25/2025).
- [36] Sruti Das Choudhury et al. “Holistic and Component Plant Phenotyping Using Temporal Image Sequence”. In: *Plant Methods* 14.1 (May 10, 2018), p. 35. ISSN: 1746-4811. DOI: 10.1186/s13007-018-0303-x. URL: <https://doi.org/10.1186/s13007-018-0303-x> (visited on 05/26/2025).
- [37] Esmael Hamuda, Martin Glavin, and Edward Jones. “A Survey of Image Processing Techniques for Plant Extraction and Segmentation in the Field”. In: *Computers and Electronics in Agriculture* 125 (July 2016), pp. 184–199. ISSN: 01681699. DOI: 10.1016/j.compag.2016.04.024. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0168169916301557> (visited on 05/26/2025).
- [38] Zhibin Wang et al. “Image Segmentation of Overlapping Leaves Based on Chan–Vese Model and Sobel Operator”. In: *Information Processing in Agriculture* 5.1 (Mar. 1, 2018), pp. 1–10. ISSN: 2214-3173. DOI: 10.1016/j.inpa.2017.09.005. URL: <https://www.sciencedirect.com/science/article/pii/S2214317317301270> (visited on 05/26/2025).
- [39] Suraj Gampa. “A Data-driven Approach for Detecting Stress in Plants Using Hyperspectral Imagery”. University of Nebraska-Lincoln, 2019.
- [40] Yuhao Chen et al. “Leaf Segmentation by Functional Modeling”. In: *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). Long Beach, CA, USA: IEEE, June 2019, pp. 2685–2694. ISBN: 978-1-7281-2506-0. DOI: 10.1109/CVPRW.2019.00326. URL: <https://ieeexplore.ieee.org/document/9025489/> (visited on 05/26/2025).
- [41] J. Praveen Kumar and S. Domnic. “Image Based Leaf Segmentation and Counting in Rosette Plants”. In: *Information Processing in Agriculture* 6.2 (June 2019), pp. 233–246. ISSN: 22143173. DOI: 10.1016/j.inpa.2018.09.005. URL: <https://linkinghub.elsevier.com/retrieve/pii/S2214317318301562> (visited on 05/26/2025).
- [42] Jean-Michel Pape and Christian Klukas. “Utilizing Machine Learning Approaches to Improve the Prediction of Leaf Counts and Individual Leaf Segmentation of Rosette Plant Images”. In: *Proceedings of the Proceedings of the Computer Vision Problems in Plant Phenotyping Workshop 2015*. Proceedings of the Computer Vision Problems in Plant Phenotyping Workshop 2015. Swansea: British Machine Vision Association, 2015,

- pp. 3.1–3.12. ISBN: 978-1-901725-55-1. DOI: 10.5244/C.29.CVPPP.3. URL: <http://www.bmva.org/bmvc/2015/cvppp/papers/paper003/index.html> (visited on 05/26/2025).
- [43] Alwaseela Abdalla et al. “Fine-Tuning Convolutional Neural Network with Transfer Learning for Semantic Segmentation of Ground-Level Oilseed Rape Images in a Field with High Weed Pressure”. In: *Computers and Electronics in Agriculture* 167 (Dec. 2019), p. 105091. ISSN: 01681699. DOI: 10.1016/j.compag.2019.105091. URL: <https://linkinghub.elsevier.com/retrieve/pii/S0168169919317107> (visited on 05/26/2025).
 - [44] Daoliang Li et al. “High-Throughput Plant Phenotyping Platform (HT3P) as a Novel Tool for Estimating Agronomic Traits From the Lab to the Field”. In: *Frontiers in Bioengineering and Biotechnology* 8 (Jan. 13, 2021), p. 623705. ISSN: 2296-4185. DOI: 10.3389/fbioe.2020.623705. URL: <https://www.frontiersin.org/articles/10.3389/fbioe.2020.623705/full> (visited on 05/24/2025).
 - [45] Jose C. Tovar et al. “Raspberry Pi-Powered Imaging for Plant Phenotyping”. In: *Applications in Plant Sciences* 6.3 (2018), e1031. ISSN: 2168-0450. DOI: 10.1002/aps3.1031. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/aps3.1031> (visited on 05/24/2025).
 - [46] Christian Bauckhage and Kristian Kersting. “Data Mining and Pattern Recognition in Agriculture”. In: *KI - Künstliche Intelligenz* 27.4 (Nov. 1, 2013), pp. 313–324. ISSN: 1610-1987. DOI: 10.1007/s13218-013-0273-0. URL: <https://doi.org/10.1007/s13218-013-0273-0> (visited on 05/25/2025).
 - [47] Amine Saddik et al. “Mapping Agricultural Soil in Greenhouse Using an Autonomous Low-Cost Robot and Precise Monitoring”. In: *Sustainability* 14.23 (23 Jan. 2022), p. 15539. ISSN: 2071-1050. DOI: 10.3390/su142315539. URL: <https://www.mdpi.com/2071-1050/14/23/15539> (visited on 05/24/2025).
 - [48] Dawei Li et al. “Digitization and Visualization of Greenhouse Tomato Plants in Indoor Environments”. In: *Sensors* 15.2 (2 Feb. 2015), pp. 4019–4051. ISSN: 1424-8220. DOI: 10.3390/s150204019. URL: <https://www.mdpi.com/1424-8220/15/2/4019> (visited on 05/24/2025).
 - [49] Yonghyun Kim, Jinyoung Y. Barnaby, and Scott E. Warnke. “Development of a Low-Cost Automated Greenhouse Imaging System with Machine Learning-Based Processing for Evaluating Genetic Performance of Drought Tolerance in a Bentgrass Hybrid Population”. In: *Computers and Electronics in Agriculture* 224 (Sept. 1, 2024), p. 108896. ISSN: 0168-1699. DOI: 10.1016/j.compag.2024.108896. URL: <https://www.sciencedirect.com/science/article/pii/S0168169924002874> (visited on 05/24/2025).

- [50] Massimo Minervini, Hanno Scharr, and Sotirios A. Tsaftaris. “Image Analysis: The New Bottleneck in Plant Phenotyping”. In: *IEEE Signal Processing Magazine* 32.4 (July 2015), pp. 126–131. ISSN: 1053-5888, 1558-0792. DOI: 10.1109/MSP.2015.2405111. URL: <https://ieeexplore.ieee.org/document/7123050/> (visited on 05/24/2025).
- [51] Rafael Massahiro Yassue et al. “A Low-cost Greenhouse-based High-throughput Phenotyping Platform for Genetic Studies: A Case Study in Maize under Inoculation with Plant Growth-promoting Bacteria”. In: *The Plant Phenome Journal* 5.1 (Jan. 2022), e20043. ISSN: 2578-2703, 2578-2703. DOI: 10.1002/ppj2.20043. URL: <https://acsess.onlinelibrary.wiley.com/doi/10.1002/ppj2.20043> (visited on 05/24/2025).